

AIR: warm or humid air = LESS dense → rises → LOW (rain)

cold + dry air = MORE dense → sinks → HIGH (clear, dry)

Wind blows HIGH → LOW · densest air = POLES (cold + dry)

0° **L** ITCZ WET · 30° **H** DRY · 60° **L** WET · 90° **H** DRY

Cells: Hadley 0–30° · Ferrel 30–60° · Polar 60–90°

Winds: trades 0–30° · westerlies 30–60° · polar east. 60–90°

Coriolis: RIGHT in NH, LEFT in SH · 0 at equator, ↑ w/ lat.

↑ w/ speed + mass · big scale only · does NOT cause wind

NH: air flowing into a LOW veers right → COUNTERCLOCKWISE

Land heats + cools FASTER than water → breezes, monsoons

Sea breeze = DAY (sea → land) · land breeze = NIGHT

Monsoon = SEASONAL; summer: ocean → land = WET

★ Hurricane needs: water > 27°C, low pressure, warm air w/

high water vapor, low wind shear (< 20 knots), Coriolis

Fuel = LATENT HEAT released when water vapor condenses

Hazards: surge, wind, rain, flooding, waves, landslides

Surge ↑: strong onshore wind, big storm, head-on, wide shelf

Path: steered AROUND highs (CW) → W, then N, then NE

OCEAN: surface currents = WIND · 10% of ocean · FAST

thermohaline = DENSITY (gravity) · 90% · SLOW

Thermocline: 200 m (13°C) → 1000 m (4°C) = mixing barrier

Ekman: surface 45°, NET transport 90° RIGHT of wind (NH)

Gyres: CLOCKWISE in NH, COUNTERCLOCKWISE in SH

W boundary (Gulf Stream) = FAST, narrow, deep, WARM

why: Earth rotates E + Coriolis ↑ toward poles → squeezed

Upwelling = cold, nutrient-rich water up → plankton → fish

NH W coast: wind from N → offshore → UPWELLING

gyre center converges → downwelling · equator upwells

★ El Niño: FEWER Atlantic hurricanes (↑ shear), MORE Pacific

★ La Niña: MORE Atlantic hurricanes (↓ shear, weak trades)

Deep water forms at poles: cold + salty (sea ice) → SINKS

Fresh meltwater → conveyor SLOWS (Younger Dryas cooling)

Topic → page**ATMOSPHERIC CIRCULATION**

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OCEAN CIRCULATION

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Legend: ★ = likely exam Q · red = key idea · → = leads to

NH/SH = N/S hemisphere · H/L = high/low pressure

SST = sea-surface temp · ↑ ↓ = increase / decrease

Exam Wed 9/30 · on paper · INDIVIDUAL part (30 min) first,
then TEAM part (20 min)

★ Qs come from CLASS ASSIGNMENTS too → review them (p. 15)

ATMOSPHERIC CIRCULATION (The Atmosphere + Wind)

Why care? wind starts ocean circulation · ocean + air trade

heat + gases · storms reshape the coast

1. Air Density — how + why it changes

Air moves 2 ways:

KEY HORIZONTAL (along surface) → PRESSURE: flows HIGH → LOW

VERTICAL → DENSITY: less dense air RISES, dense air sinks

+ Earth's rotation (Coriolis) changes the direction

Sun = ultimate energy source for ocean + wind circulation

★ Air density is controlled by 2 things:

1) TEMPERATURE: warm air → molecules spread out → fewer per volume → LESS dense → RISES

cold air → molecules packed close → MORE dense → SINKS

2) MOISTURE: water vapor H₂O (18) is LIGHTER than

N₂ (28, 78% of air) + O₂ (32, 21% of air)

★ trick → humid air is LESS dense than dry air → rises

∴ Densest air = POLES (cold + dry)

least dense = equator (warm + humid)

2. High + Low Pressure — why they form + weather

WHY WHY = DIFFERENTIAL HEATING of Earth:

equator: sun overhead → energy concentrated → HOT

poles: sun low in sky → same beam spread out → COLD

→ heat SURPLUS at low latitudes, DEFICIT near poles

→ atmosphere + oceans move surplus heat toward poles

(seasons: Earth's tilt changes where sun is most direct)

Density changes start air moving → make H + L regions

LOW pressure (L)

warm, humid, less dense air

air RISES + COOLS → condenses

CLOUDS, RAIN, storms

surface winds flow IN

where: 0° (ITCZ) + 60°

HIGH pressure (H)

cold, dry, dense air

air SINKS + WARMS → no clouds

CLEAR, DRY, calm

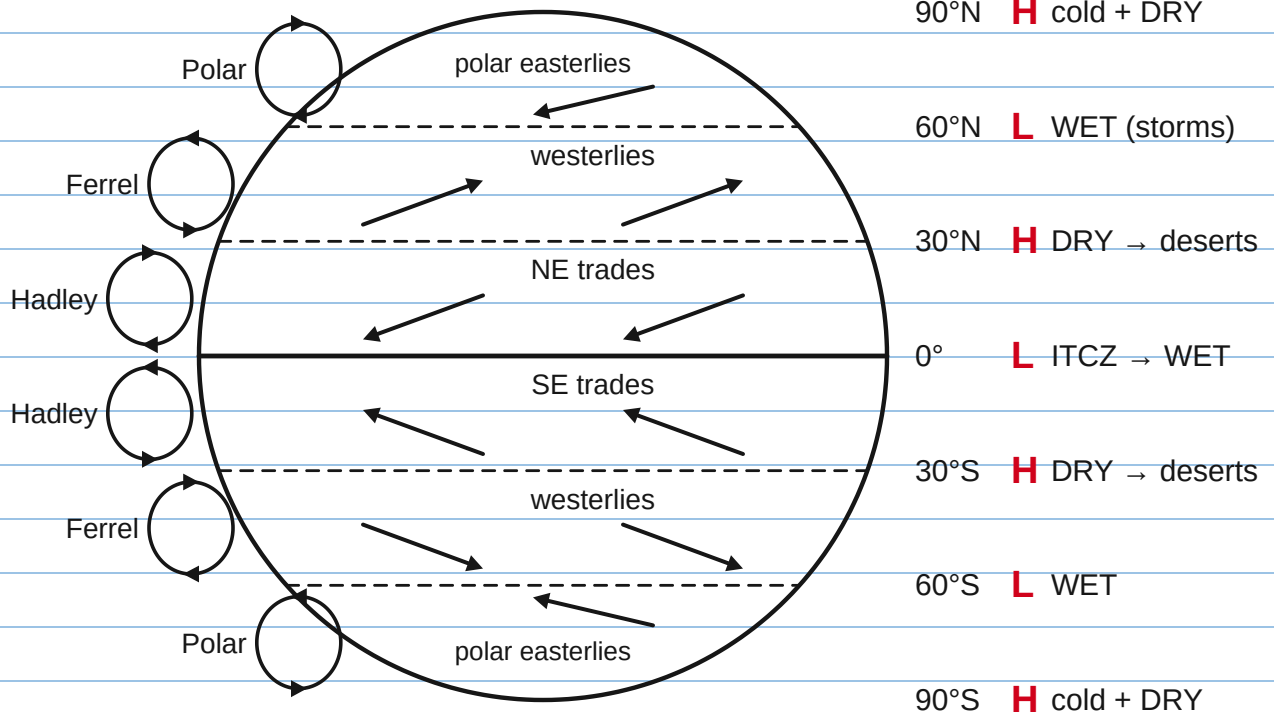
surface winds flow OUT

where: 30° + 90° (poles)

3. Global Atmospheric Circulation ★ (DRAW IT)

Non-rotating Earth → 1 big cell per hemisphere

Rotating Earth (Coriolis) → 3 CELLS per hemisphere:

**ITCZ** 0° = ITCZ (Intertropical Convergence Zone) / doldrums:

NE + SE trades meet → air RISES → LOW → heavy rain

30° = horse latitudes / subtropical HIGHS: dry air sinks

→ world's DESERTS (Sahara, Arabia, Kalahari, Australia)

60° = subpolar LOWS: air rises → rain/snow, storms

90° = polar HIGHS: cold dry air sinks → very dry

ITCZ follows the sun: July = farther N, January = farther S

Winds are named for where they come FROM (NE trades)

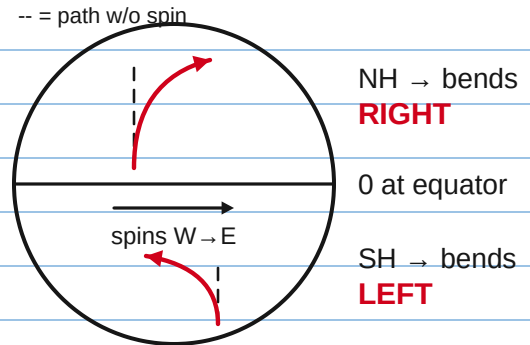
4. Coriolis Effect ★**DEF** = APPARENT deflection of objects moving across Earth**WHY** WHY: Earth rotates W → E; every latitude makes 1 turn a day

but the equator travels farther → moves FASTER

(equator ~40,000 km/day · 60° ~20,000 km/day · pole 0)

- object moving N/S keeps its OLD eastward speed
- ground under it is faster/slower → path CURVES

- ★ **RESULT:** deflects to RIGHT of travel in NH, LEFT in SH
- NONE at equator, ↑ with latitude
- ↑ with SPEED + MASS of object
- Only big motions (> tens of km)
- Does NOT cause wind — only changes its DIRECTION



Ex: wind H → L gets bent right (NH) · makes gyres + storms spin (NH: air into a LOW veers right → COUNTERCLOCKWISE)

5. Monsoons

DEF = SEASONAL reversal of winds → WET season + DRY season

WHY WHY: land heats + cools FASTER than water (water has a high heat capacity → takes lots of heat to change temp)

SUMMER: land hotter → air rises over land → LOW

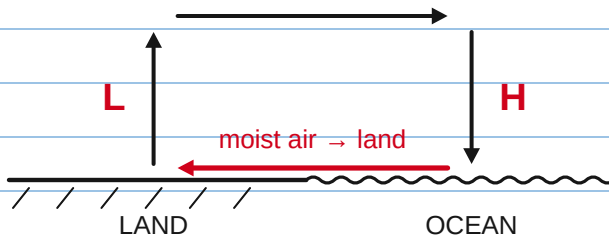
→ moist ocean air flows ONTO land → rises, cools → RAIN

WINTER: land colder → air sinks over land → HIGH

→ dry air flows from land OUT to sea → DRY season

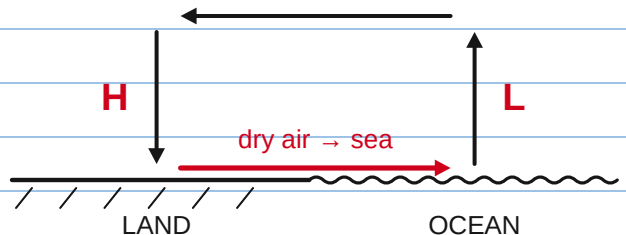
SUMMER → WET season

land HOTTER than ocean



WINTER → DRY season

land COLDER than ocean



Moves w/ ITCZ: Jun–Jul ITCZ N over India (Kozhikode rain)

Dec–Jan ITCZ S → rain at Darwin, Australia

6. Land + Sea Breeze

Same idea as monsoon but DAILY + small (coasts only)

DAY: land heats faster → warm air rises over land (L)

→ cool air sinks over sea (H) → ONSHORE = SEA BREEZE

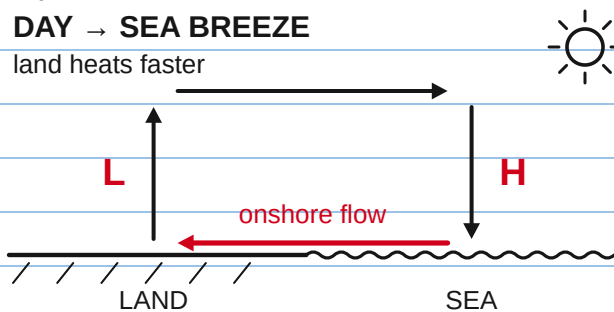
NIGHT: land cools faster → sea is warmer → rises over sea

(L), sinks over land (H) → OFFSHORE = LAND BREEZE

TIP Tip: a breeze is named for where it comes FROM

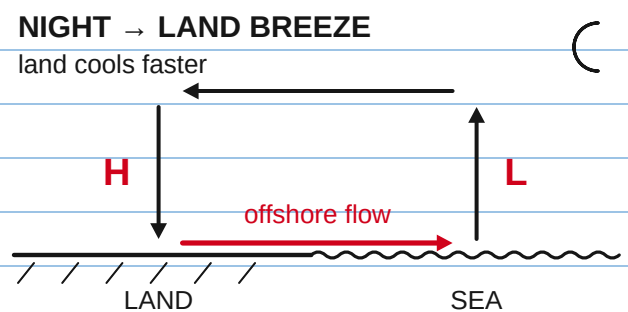
DAY → SEA BREEZE

land heats faster



NIGHT → LAND BREEZE

land cools faster



7. What a Hurricane Needs to Form ★

tropical cyclone = hurricane (Atl.) = typhoon (W Pacific)

★ **LIST** **NEED** (class list):

- 1) WARM WATER $> 27^{\circ}\text{C}$ → heat + moisture = fuel
- 2) a LOW-PRESSURE system to start from
- 3) WARM air w/ HIGH WATER VAPOR
- 4) LOW WIND SHEAR (< 20 knots) → storm stays upright
(high shear TILTS the storm + tears it apart)
- 5) STRONG CORIOLIS → spin (none at equator)

WHY **ENGINE = LATENT HEAT:** warm ocean evaporates water (takes in heat) → air rises, vapor condenses → RELEASES heat

→ air warms, rises faster → pressure drops → stronger

Cyclonic flow: air rushes toward the L, veers RIGHT (NH)

→ spins COUNTERCLOCKWISE in NH (clockwise in SH)

Structure: moist surface winds spiral IN → rise in eyewall

(winds > 250 km/h) → exit at top · calm EYE ~13–16 km

Form over warm tropical water (not at equator, none in

S Atlantic) · die over land or cold water (no fuel)

8. Hurricane Hazards (Risks)

LIST WAVES · WIND · RAIN · FLOODING · LANDSLIDES · STORM SURGE

Rain flooding can be far inland (Ivan → Asheville NC, 2004)

★ **STORM SURGE** = sea pushed up onto the coast

caused by strong ONSHORE WINDS (wind-driven)

low pressure lifts water a little (only ~5% of surge)

water "piles up" as it reaches shallow water at coast

worst if it hits at HIGH TIDE (Sandy 2012, Atlantic City)

★ factor	surge is HIGHER when...
wind speed	winds are STRONGER
radius of storm	storm is BIGGER
forward speed	fast = higher on open coast slow = floods longer + wider
angle of approach	hits PERPENDICULAR (head-on)
continental shelf	WIDE + gentle slope (shallow)
shape of coastline	bays / inlets funnel water in

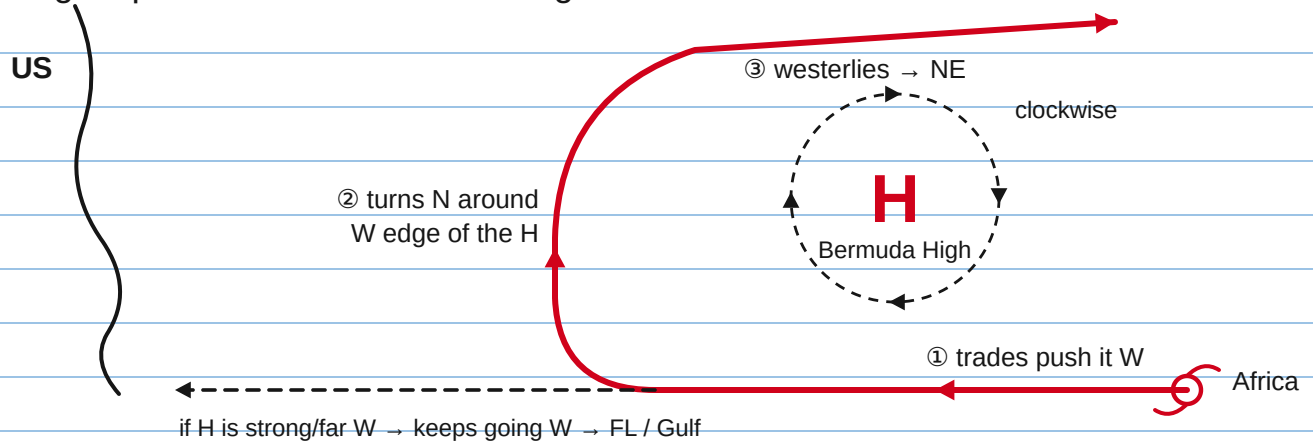
9. Predicting a Hurricane's Path ★

Hurricanes are STEERED by winds around H + L systems

Wind flows H → L · storm can't push INTO a high → goes

AROUND its edge (CLOCKWISE around a H in NH)

+ gets pulled toward LOWS / troughs



Typical Atlantic track: ① forms off Africa, trades push W

② reaches W edge of Bermuda High → turns N

③ meets westerlies (~30°N) → curves NE, out to sea

Strong High stretching W → storm keeps going W → FL/Gulf

Weak High / High shifted E → turns N early → out to sea

or up the East Coast · LOW over E US pulls storm N

10. Hurricanes + Climate Change**FUTURE**

EXPECTED as climate warms:

- warmer SST = more fuel → STRONGER storms (more Cat 4–5)
- warmer air holds more vapor → MORE RAIN + flooding
- sea-level rise → HIGHER STORM SURGE, more coast flooded
- faster intensification · may reach farther from tropics
- total # of storms: same or FEWER (no big increase)

PAST

OBSERVED over last ~100 yrs:

- total # of hurricanes: NO clear long-term trend (old records miss storms at sea — no satellites pre-1960s)
- ↑ share of MAJOR (Cat 3–5) storms + ↑ rainfall
- \$ damage ↑ a lot → more people + buildings on coasts

**NOT more hurricanes, but STRONGER + WETTER ones**

OCEAN CIRCULATION

Why care? currents regulate CLIMATE (move + store heat, store gases) + control PRODUCTIVITY (nutrients, O₂)

Key concepts: gyres · Ekman · W boundary currents · upwelling/downwelling · conveyor belt

11. Surface vs Deep Water ★

Ocean is LAYERED by density (cold + salty = dense)

DEF THERMOCLINE = layer where temp drops FAST with depth:
~200 m (660 ft) 13°C / 55°F → ~1000 m (3300 ft) 4°C / 39°F
= BARRIER → surface + deep water don't mix

	SURFACE layer	DEEP layer
temp	WARM (varies)	COLD (~4°C or less)
density	LOW	HIGH
mixing	mixed by WIND + waves	not mixed by wind
light	sunlit → photosynthesis	dark
nutrients	LOW (used by plankton)	HIGH (from decay)
oxygen	HIGH (air + photosyn.)	lower
moved by	WIND · 10% · FAST	DENSITY · 90% · SLOW
salinity	varies (evap. vs rain)	uniform

Salinity = parts per thousand (ppt, ‰) · avg ~35 (34.7)

salt comes from erosion of continents + hydrothermal vents

HIGH in subtropics (23.5–35°): evaporation > rain (dry

sinking air at 30°) · Atlantic = saltiest ocean

LOW at equator (rain), near rivers (Amazon, Ganges)

12. Major Ocean Currents (recognize them)

4 things Controlled by: WIND · DENSITY · CONTINENTS · CORIOLIS

★ GYRES = big loops: CLOCKWISE in NH, COUNTERCLOCKWISE SH

5 gyres: N + S Atlantic, N + S Pacific, Indian Ocean

equatorial currents flow W (trades) · ~40–50° flow E

W boundary = warm, flows toward POLE · E boundary = cold, flows toward EQUATOR

gyre	W boundary (WARM)	E boundary (COLD)
N Atlantic	Gulf Stream	Canary
S Atlantic	Brazil	Benguela
N Pacific	Kuroshio	California
S Pacific	East Australian	Peru (Humboldt)
Indian	Agulhas	West Australian

Also: N Atlantic Current (→ Europe) · N Pacific Current

N + S Equatorial Currents (→ W) · Eq. Countercurrent (→ E)

Antarctic Circumpolar Current (West Wind Drift) → E

all the way around Antarctica · cold: Labrador, Oyashio

Gulf Stream: seen 1513 (Ponce de León), charted 1770s

by Ben Franklin · flow $\approx 300\times$ the Amazon River

13. Forces That Drive Surface + Deep Currents ★

	SURFACE currents	DEEP (thermohaline)
driven by	WIND (drag on surface)	DENSITY → GRAVITY
amount	~10% of ocean	~90% of ocean
speed	FAST	SLOW
where	above thermocline	below thermocline

Surface currents also turned by CORIOLIS + CONTINENTS

EKMAN SPIRAL: wind drags top layer, each layer drags the

one below → each deeper layer SLOWER + turned more right

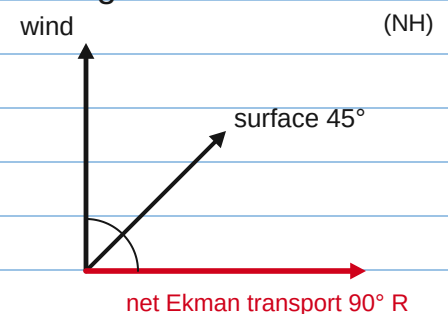
surface water moves 45° RIGHT

of the wind (NH)

★ NET water movement = EKMAN

TRANSPORT = 90° RIGHT of wind

in NH · 90° LEFT in SH



DEEP: THERMO (temp) + HALINE (salt) → density

density is "fixed" at the SURFACE, then water sinks

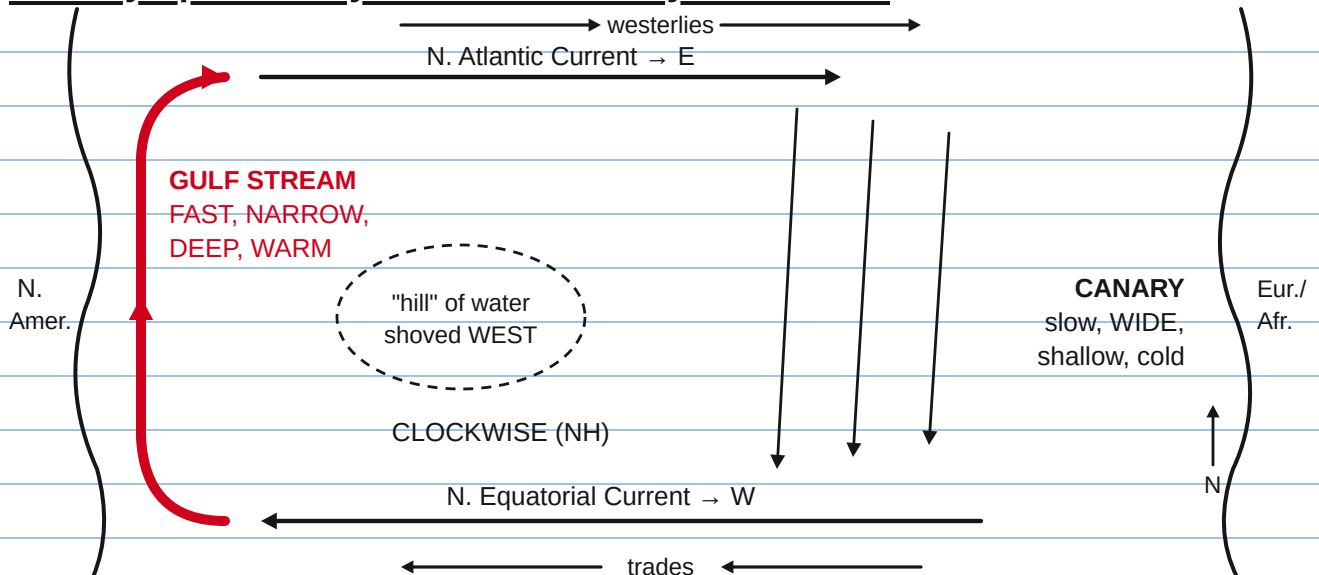
temp: cooled/heated by contact w/ the atmosphere

salinity ↑ evaporation + SEA ICE forming (leaves salt)

salinity ↓ precipitation, rivers, melting ice

cold + salty = DENSEST → sinks near the POLES

14. Why Speeds Vary: W vs E Boundary Currents ★



	WESTERN boundary	EASTERN boundary
speed	FAST	SLOW
width	NARROW	WIDE
depth	DEEP	SHALLOW
temp	WARM (from equator)	COLD (from poles)
flow	Gulf Stream ~55 Sv	Canary ~16 Sv

★ WHY WHY = western intensification, caused by:

- 1) EASTWARD rotation of Earth
- 2) Coriolis INCREASES with distance from equator
→ these COMPRESS the W boundary current → same water squeezed into a narrow path → SPEED ↑
(Sv = Sverdrup = 1 million m³ of water / sec)

15. How Surface Currents Affect Weather

Currents carry HEAT from equator → poles

WARM currents (W boundary, E coasts) → warm, humid air,
lots of evaporation → RAIN + fuel for storms/hurricanes

COLD currents (E boundary, W coasts) → cool air, fog,
little evaporation → DRY coasts (Atacama, Namib deserts)

EX Ex: Gulf Stream + N Atlantic Current warm W Europe:

Jan avg high/low: Newfoundland 30/16°F vs Ireland 46/37°F
(about the same latitude!)

Water's high heat capacity → coasts milder than inland

16. Upwelling + Downwelling ★

DEF UPWELLING = deep, cold, NUTRIENT-RICH water rises

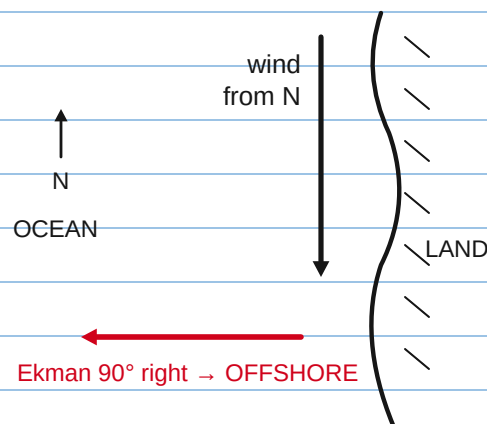
DEF DOWNWELLING = surface water piles up + SINKS

Coastal, NH west coast (e.g. California):

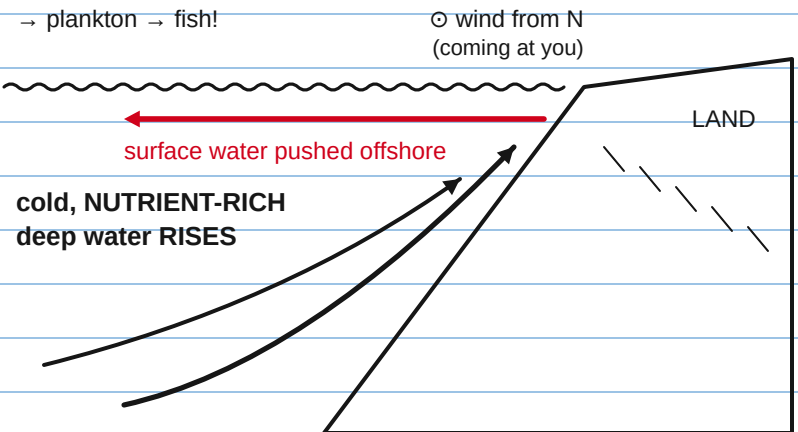
wind from NORTH → Ekman 90° right → water moves OFFSHORE
→ deep water rises to replace it = UPWELLING

wind from SOUTH → water moves ONSHORE → piles up at
coast → sinks = DOWNWELLING

MAP VIEW (NH west coast)



SIDE VIEW



EQUATORIAL: Coriolis turns water RIGHT north of equator +
LEFT south of it → water DIVERGES → UPWELLING

SOUTHERN OCEAN: winds around Antarctica → upwelling

GYRE CENTER: water CONVERGES → DOWNWELLING

★ MARINE LIFE:

upwelling → nutrients into sunlight → phytoplankton →
zooplankton → FISH · best fisheries: California, Peru,
Canary, Benguela (all EASTERN boundary currents)
downwelling → carries O₂ down to deep-sea life, but
surface stays nutrient-poor → low productivity

CLIMATE: upwelled cold water → cool, foggy coast · mixing
moves heat + gases (O₂, CO₂) between surface + deep

17. El Niño + La Niña (ENSO) ★

ENSO = El Niño–Southern Oscillation (air-pressure seesaw

between W + E Pacific) · ANOMALY = difference from avg

NORMAL: trade winds blow E → W → push warm water WEST

→ warm pool + LOW (rain) over Indonesia/Australia

→ cold UPWELLING off Peru (HIGH, dry, great fishing)

★ EL NIÑO: trades WEAKEN → warm water sloshes EAST

→ thermocline deepens in E → upwelling stops → fish die

→ RAIN + FLOODS in Peru · DROUGHT in Indonesia/Austr.

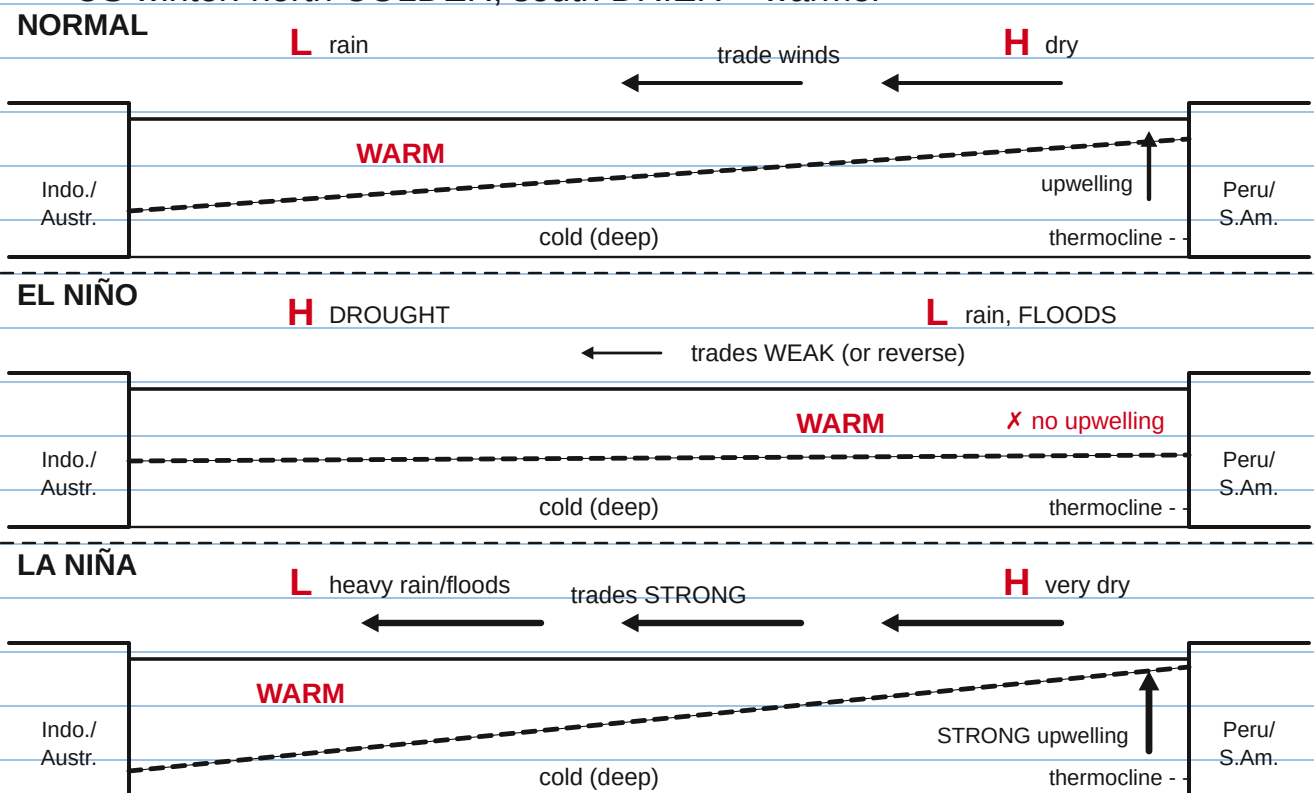
→ US winter: north WARMER, south WETTER

★ LA NIÑA: trades STRONGER → even more warm water W

→ E Pacific colder than normal, strong upwelling

→ floods in Indonesia/Australia · dry Peru

→ US winter: north COLDER, south DRIER + warmer



★ HURR.	ATLANTIC hurricanes	E + central PACIFIC
El Niño	FEWER: ↑ shear + trades, stable	MORE: ↓ wind shear
La Niña	MORE: ↓ shear + weak trades	FEWER: ↑ wind shear

18. Deep Water Circulation + Climate ★

Deep water FORMS at poles: cooling + sea ice (salty) → dense → SINKS → spreads along bottom → slowly rises

AABW = Antarctic Bottom Water (densest, Weddell Sea)

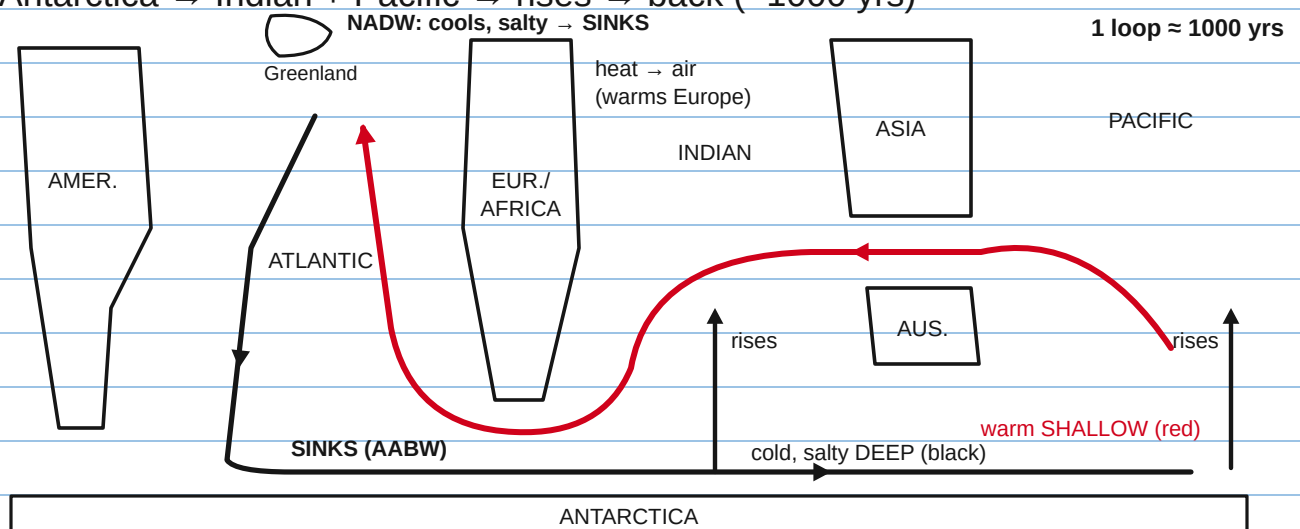
NADW = North Atlantic Deep Water (near Greenland)

Pacific deep water: weak layers, sluggish, uniform

CONVEYOR BELT: warm shallow current → N Atlantic → gives

heat to air → cold + salty → SINKS → deep current S →

Antarctica → Indian + Pacific → rises → back (~1000 yrs)

**★ CLIMATE EFFECTS:**

moves heat equator → poles → keeps N Europe mild

carries O₂ + CO₂ (dissolved gases) into the deep ocean

IF IT SLOWS: melting ice + rain → FRESHER, less dense water

→ can't sink → conveyor slows → less heat → N. cools

Happened before: YOUNGER DRYAS (~12,900–11,500 yrs ago)

Lake Agassiz meltwater → NH back to near-glacial cold

· ended fast: Greenland +10°C in about a decade

Today: AMOC slowed in 20th century → "cold blob" S of

Greenland cooled while the rest of Earth warmed

Class assignments / activities